stream is low, and Coriolis force is weak leading to meandering) with latitude.

- Rossby waves are formed when polar air moves toward the Equator while tropical air is moving poleward.

79. **Statement 1: INCORRECT**—The sub-tropical jet stream is produced by the earth's rotation (Coriolis force) and temperature contrast between tropical and sub-tropical regions.
During winter, the STJ is nearly continuous in both hemispheres. The STJ exists all year in the southern hemisphere. However, it is intermittent in the northern hemisphere during summer when it migrates north.
Statement 2: CORRECT—The polar front jet is produced by a temperature difference and is closely related to the polar front.
Statement 3: CORRECT—It has a more variable position than the sub-tropical jet. In summer, its position shifts towards the poles and in winter towards the equator.

80. **Statement 1: INCORRECT**—A cold air mass is one which is colder than the underlying surface and is associated with instability and atmospheric turbulence.
Statement 2: INCORRECT—A warm air mass is one which is warmer than the underlying surface and is associated with stable weather conditions.

81. **Statement 1: CORRECT**—The temperature contrast influences the thickness of frontal zone in an inversely proportional manner, i.e., two air masses with higher temperature difference do not merge readily. So the front formed is less thick.
Statement 2: CORRECT—The frontal activity is invariably associated with cloudiness and precipitation because of ascent of warm air which cools down adiabatically, condenses and causes rainfall.

82. **Statement 1: CORRECT**—No cumulonimbus clouds are formed along warm front as the gradient is gentle.
Statement 2: CORRECT—Warm Front: It is a sloping frontal surface along which active movement of warm air over cold air takes place (warm air mass is too weak to beat the cold air mass).
Frontolysis (front dissipation) begin when the warm air mass makes way for cold air mass on the ground, i.e. when the warm air